



Department of Computer Science and Engineering
Jashore University of Science & Technology
3rd Year 2nd Semester Final Examination-2024

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Course Code: CSE 3201

Course Title: Artificial Intelligence and Machine Learning

Full marks: 72

Time: 3.00 hours

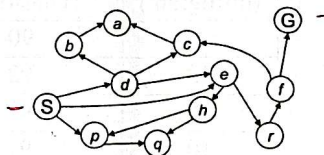
[Figures in the right margin indicate full marks. Answer any **SIX** questions]

1. a) Define Artificial Intelligence (AI) and explain its main types with examples. 3
 - b) What things AI can do in today's world? 2
 - c) When performing an action, how a reflex agent and a planning agent will behave? 2
Explain with example. 4
 2. a) What is a search problem in AI and what are the main components of a search problem? 3
 - b) Explain state space graph and search tree in a search problem. 3
 - c) Consider a Tic-Toe game with 6
 - Agent positions: 120
 - Food count: 30
 - Ghost positions: 12
 - Agent facing: NSEW

Now, calculate the followings

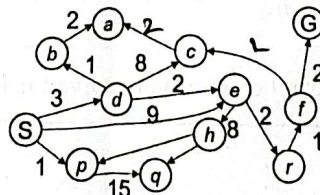
- Number of World states
- Number of states for pathing
- Number of states for eat-all-dots

- 3) a) Apply Depth-First Search (DFS) on the following example



- b) Answer the following questions for Breadth-First Search (BFS) 3
- (i) How much space does the fringe take?
 - (ii) Is it complete?
 - (iii) Is it optimal?

4. a) What is the difference between uninformed search and informed search? 3
b) Apply UCS on the following example 9



$\sigma\left(\frac{-1}{1+\tilde{\sigma}^n}\right) \cdot \left(\sigma\left(\frac{-x}{2^n}\right)\right) \cdot \left(\frac{1}{(1+\tilde{\sigma}^n)} - \frac{1}{(1+\tilde{\sigma}^n)^2}\right) \sigma(m) - \sigma(m)^2$

* ~~5~~ a) What are the main differences between supervised, unsupervised, and reinforcement learning? Provide one example for each type. 4

4 b) A mechanical engineer has sent you the following email message. Prepare a detailed reply. "You may remember that yesterday we met in world scientific conference. As I mentioned, I am working on devising new composite materials. I would like to try using the K-Nearest Neighbours (K-NN) algorithm to build predictive models for classifying the material is thermally stable or not based on some features of material. But I don't know how does K-NN work? Can you describe the working process of K-NN algorithm and give me some advice on how to select the value of K in the K-NN Algorithm?" 5

3 c) What is overfitting in machine learning? How can it be prevented? 3

~~6~~ a) What is Machine Learning? Why is the Machine Learning trend emerging so fast? What are the application areas of machine learning in modern life? 4

4 b) Describe different types of machine learning algorithms with example. 4

2 c) How machine learning is different from general programming? 2

2 d) Why we cannot use linear regression for a classification task? 2

~~7~~ a) Suppose, medical researchers hired you to understand the relationship between drug dosage and blood pressure of patients. What kind of machine learning problem is it? 2

b) You are given the following data samples for the above-mentioned problem. Consider a linear model for it. What will be the training losses when $W=0,1,2,3$, and 4 in case of MSE loss? 6

Dosage (milligram)	Blood Pressure (Diastolic)
5	90
13	92
19	93
10	91

3 e) What is learning rate in Machine learning? Discuss the problems associated with learning rate and how can we choose a better learning rate? 4

~~8~~ a) Suppose, we have a dataset that has two features for each data sample. 10

i) Design a linear model to solve this problem and draw the computational graph. 10

12 ii) Assume that the two learnable parameters are W_1 and W_2 , and the loss function is MSE. Find $\frac{\partial Loss}{\partial W_1}$ and $\frac{\partial Loss}{\partial W_2}$ when $X_1=2$, $X_2=3$, $Y=7$. Initialize W_1 and W_2 with 3 and 2, respectively. 12

2 b) What types of problems can be effectively solved using unsupervised learning methods? 2



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Semester Final Examination-2024
3rd Year 2nd Semester, Session: 2020-2021

Time: 3.00 hours
Marks: 72

Course Code: CSE 3203

Course Title: Compiler Design and Automata Theory

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

57
72

1. a) Differentiate between a compiler and an interpreter in terms of execution speed, error detection, and use cases. 04

b) What is code optimization in the context of compilers? Explain its significance and provide an example of machine-independent optimization. 04

c) For the expression $x = (a + b) * (c - d)$: 04

I. Write the three-address code sequence.

II. Explain how the three-address code helps in intermediate code generation.

2. a) Explain the differences between NFA and DFA with respect to: 06

I. Determinism

II. State complexity

III. Transition representation.

Provide examples where an NFA is more compact than the corresponding DFA.

b) Construct the NFA for all binary strings where the second last bit is 1. Then, convert this NFA into an equivalent DFA and explain the conversion process. 06

3. a) Discuss syntax error handling techniques. 02

b) Design grammars for the following languages. 4*2 = 08

I. The set of all strings of 0s and 1s such that every 0 is immediately followed by at least one 1.

II. The set of all strings of 0s and 1s that are palindromes; that is, the string reads the same backward as forward.

III. The set of all strings of 0s and 1s with an equal number of 0s and 1s.

IV. The set of all strings of 0s and 1s in which 011 does not appear as a substring.

c) Explain the dangling else problem. 02

4. Consider the following grammar

$lexp \rightarrow atom \mid list$

$atom \rightarrow number \mid identifier$

$list \rightarrow (lexp_seq)$

$lexp_seq \rightarrow lexp_seq \ lexp \mid lexp$

a) Re-write the grammar by eliminating left recursion. 03

b) Construct the FIRST and FOLLOW sets for the new grammar. 05

c) Construct the LL(1) parsing table for the resulting grammar.

04

5. a) Construct the regular expression for recognizing an *identifier* in C.

02

b) Draw the transition diagram for an *identifier* in C.

03

c) Divide the following C code snippet into appropriate lexemes. Identify each token type.

05

```
int main() { int x = 10; printf("Hello, world!\n"); return 0; }
```

d) Compare between compiler and interpreter.

02

6. a) Suppose you are taking input in your website. You need to validate if a user is providing a valid email address. Construct the regular expression for that. The considerations you need to take into account are:

02

- There must be some alphanumeric characters before the @ symbol.
- The email address must end with .com or .com.bd or .com.au
- Between @ and .com, there must be some alphanumeric characters.

b) Create the NFA for the regular expression $((a^*b)(ba))^*a$ step by step.

04

c) Construct the DFA for the NFA resulting from question 6.b.

06

7. a) Consider the context-free grammar:

$S \rightarrow SS+ | SS* | a$

1+

and the string $aa + a^*$.

1+

I. Give a leftmost derivation for the string.

2

II. Give a rightmost derivation for the string.

III. Give a parse tree for the string.

b) Consider the grammar G at the right hand side.

$S \rightarrow AB$

2x4

$A \rightarrow aa | ab | ba | bb$

= 08

$B \rightarrow aBa | bBb | C$

$C \rightarrow aa | ab | ba | bb$

Which of the following strings are valid according to G?

i) bababbab

ii) abaab

iii) aaabbbba

iv) babaa

8. a) Differentiate between NFA and DFA.

02

b) Consider the grammar below

$S \rightarrow (L) | a$

$L \rightarrow L, S | S$

Perform Shift Reduce Parsing for the input string using the grammar.

i. (a, (a,a))

ii. (a,a)



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Semester Final Examination-2024
 3rd Year 2nd Semester, Session: 2020-2021

Time: 3.00 hours
 Marks: 72

Course Code: CSE 3205

Course Title: Peripherals, Interfacing and IoT

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

64

1. a) Define peripheral adapter. Explain how a peripheral adapter connects to the data highway with appropriate diagram. 03 (3)
- b) What are the primary goals and challenges in designing interfaces for communication between a computer and external devices or systems? 04 (2)
- 10 c) Write an assembly language program that checks the status of three devices (DEV1, DEV2, DEV3) using Programmed I/O and invokes corresponding service routines (SR1, SR2, SR3) based on their readiness. Poll the three status registers (STAT1, STAT2, STAT3) until a device is ready, and loop back if no device is ready. 05 (5)
2. a) Define Interrupt Vector Table (IVT). Explain the working principle of the daisy chain method used for handling priority interrupts with a block diagram. 04
- b) Design and describe a priority interrupt handling system with a 4-to-2 priority encoder. Draw necessary diagrams and logic table for your design. 08
3. a) What are the primary functions of a DMA controller in managing data transfers between devices and memory? 03 (3)
- 11 b) Describe the typical architecture of a DMA controller and its main components with proper block diagram. 05 (5)
- c) What are the emerging trends and advancements in DMA technology, particularly in high-speed data transfer scenarios? 04 (3)
4. a) Differentiate between Programmed I/O and Memory Mapped I/O. (2) 03
- b) Write an assembly language program that implements Programmed I/O to receive a character from a data register. Assume the address of the Status Register is 0FFH, and the Data Register is 0FEH. 04
- 10 The program should:
 - Continuously poll the Status Register to check the "Receive Ready" bit (bit 0) until it is set.
 - Once ready, read the character from the Data Register into a register (e.g., AL).
 - The character should be stored at the memory location labeled Buffer.
- c) In a system using nested interrupts, three interrupts A (highest priority), B (medium priority), and C (lowest priority) arrive with different priority levels. If Interrupt B occurs first, followed by Interrupt C and then Interrupt A, explain how the processor will handle them step by step. Draw necessary diagrams. 05 (4)
5. a) Differentiate between Single Handshake I/O and Double Handshake I/O 03
- b) Write short notes on the following data conversion methods: 04
 - i. Serial-parallel conversion
 - ii. Parallel-serial conversion

- c) Define bit stuffing. Suppose device X wants to send a data frame to device Y using High-Level Data Link Control (HDLC) format. Considering the following values and calculate the final HDLC frame to be sent. 05

Address= 11111100

Control= 00110011

Data= 11111000 11111100

Frame Check Sequence= 11001100 10101010

- 11 ~~6.~~ a) How can robust security and privacy mechanisms be developed to protect data and devices in IoT networks from cyberattacks? 04 (3)
- b) Write short notes on the following errors occurred in sensors: 03 (3)
- i. Hysteresis error ii. Quantization error iii. Aliasing error
- c) Explain the working principles of RFID (Radio Frequency Identification) and NFC (Near Field Communication) with necessary diagrams. Mention the key differences between RFID and NFC. 05 (5)
- 12 ~~7.~~ a) Define Hot Swapping. Briefly explain the four types of data transfer mechanism in USB with appropriate diagrams. 04 (4)
- b) Explain the different layers of HART (Highway Addressable Remote Transducer) protocol. Mention the advantages of WirelessHART over Zigbee. 04 (4)
- c) Explain the different operational phases of AODV (Ad-hoc On-Demand Distance Vector) protocol. Mention its advantages over LOADng (Lightweight On-demand Ad hoc Distance-vector Next Generation). 04 (4)
- 10 ~~8.~~ a) Draw a network diagram of Advanced Message Queuing Protocol (AMQP). Explain the four types of AMQP exchanges. 04 4
- b) How can blockchain technology enhance security, transparency, and data management in IoT ecosystems? 04 3
- c) Draw the different types of Zigbee network topologies. Analyze their differences in terms of communication and network performance. 04 3



Course Code: AIS 3201
Time: 3.00 hours

Course Title: Accounting and Information System
Full marks: 72

[Figures in the right margin indicate full marks. Answer any SIX questions]

1. a) What do you mean by accounting? 2
b) What are the four financial systems that are prepared by a business? What is the purpose and content of each statement? 6
c) The information that a user of financial information needs depends upon the kinds of decisions the user makes. There are two broad groups of users of financial information: internal users and external users. Who are those internal and external users of accounting information and how do they use this information? 4
2. a) Shewly decides to open a lemonade stand. She invests 100 Taka of her savings to buy lemons, sugar, and a table. She sells 120 cups lemonade for 2 Taka per cup. After a day, she counts her earnings. Explain how Shewly records her initial investment, daily sales, and calculate her profit or loss in her book of accounts. 4
b) Discuss how advancements in technology, such as blockchain and artificial intelligence, are influencing the field of accounting and financial reporting? 4
c) Explain the role of auditors in the financial reporting process. How do auditors add credibility of financial statements, and why is their independence crucial? 4
3. a) Define journalizing and posting. 2
b) Emily Valley is a licensed dentist. During the first month of the operation of her business, the following events and transactions occurred. 8
April 1 Invested Tk.20,000 cash in her business.
1 Hired a secretary-receptionist at a salary of Tk.700 per week payable monthly.
2 Paid office rent for the month Tk.1,100.
3 Purchased dental supplies on account from Dazzle Company Tk.4,000.
10 Performed dental services and billed insurance companies Tk.5,100.
11 Received Tk.1,000 cash advance from Leah Mataruka for an implant.
20 Received Tk.2,100 cash for services performed from Michael Santos.
30 Paid secretary-receptionist for the month Tk.2,800.
30 Paid Tk.2,400 to Dazzle for accounts payable due.

Instructions

Journalize the transactions.

4. a) What are major differences between financial and managerial accounting? 4
b) "The variable cost per unit varies with output, whereas the fixed cost per unit is constant." Do you agree? Explain why or why not. 4
c) Define the following terms with example: differential cost, opportunity cost, and sunk cost. 4
5. a) Define break-even point. 2
b) Menlo Company distributes a single product. The company's sales and expenses for last month follow: 10

	Total	Per Unit
	TK.	TK.
Sales	4,50,000	30
Variable Expenses	1,80,000	12
Contribution Margin	2,70,000	18
Fixed Expenses	2,16,000	
Net Operating Income	54,000	

Instructions:

- (i) What is the monthly break-even point in units sold and in sales dollars?
(ii) Without resorting to computations, what is the total contribution margin at the break-even point?

- (iii) How many units would have to be sold each month to earn a target profit of Tk. 90,000? Verify your answer by preparing a contribution format income statement at the target sales level.
- (iv) Refer to the original data; compute the company's margin of safety in both dollar and percent.
- (v) Compute the company's degree of operating leverage. Using the degree of operating leverage, estimate the impact on net operating income of a 5% increase in sales.

6. a) "A worksheet is a permanent accounting record and its use is required in the accounting cycle." Do you agree? Explain. 2
- b) Casey Hartwig, CPA, was retained by Global Cable to prepare financial statements for April 2024. Hartwig accumulated all the ledger balances per Global's records and found the following. 10

Global Cable Trial Balance		
	Debit TK.	Credit TK.
Cash	4,100	
Accounts Receivable	3,200	
Supplies	800	
Prepaid Insurance	1,200	
Equipment	9,400	
Accumulated Depreciation-Equipment		1,350
Accounts Payable		2,100
Salaries and Wages Payable		700
Unearned Service Revenue		890
Owner's Capital		12,900
Service Revenue		5,450
Salaries and Wages Expenses	3,300	
Advertising Expenses	600	
Miscellaneous Expenses	290	
Depreciation Expenses	500	
	23,390	23,390

Other data:

- (i) A count of supplies shows Tk. 300 of unused supplies on April 30.
- (ii) Prepaid insurance is a 1-year policy starting April 1, 2024.
- (iii) Two-thirds of the unearned service revenue has been earned.
- (iv) Depreciation on Equipment for the month is Tk. 205

Instructions:

Enter the trial balance on a worksheet and complete the worksheet.

7. a) "Accounting information system (AIS) is the intelligence—the information providing vehicle" Justify the statement. 2 2
- b) How does the conceptual framework of Accounting Information Systems (AIS) integrate its sub-systems, such as transaction processing systems to enhance decision-making and organizational efficiency? 6 5
- c) What are the key differences and interrelationships between Accounting Information Systems (AIS) and Management Information Systems (MIS) in terms of their objectives, components, and contributions to organizational decision-making? 4 3
8. a) How do a Database Management System (DBMS) and a traditional file system differ in terms of data storage, retrieval efficiency, data integrity, and scalability? 4 4
- b) Define DDL and DML. List the tasks under DDL and DML. 4 4
- c) What are the key principles of the Relational Database Model, and how do concepts like normalization, primary keys, and foreign keys contribute to maintaining data integrity and reducing redundancy? 4 3